

CUSTOMER CHURN ANALYSIS

Data Science & Analytics Internship

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SQL, Power BI, Microsoft Excel

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EXECUTIVE SUMMARY

This project focuses on customer churn analysis using SQL Server and Power BI to understand customer retention patterns and identify factors contributing to customer attrition. The analysis evaluates customer demographics, subscription preferences, contract types, spending behavior, and churn trends.

The dataset consists of 64,374 customer records containing demographic information, subscription details, spending behavior, and churn status. SQL was used for data querying and analysis, while Power BI was used to create an interactive dashboard for visualization.

The analysis revealed a churn rate of 47.37%, with Standard subscriptions having the largest customer base and Monthly contracts being the most common contract type. The average customer age was found to be 42 years, and the average customer spending was ₹541.

The insights generated from this project help businesses understand customer behavior, improve retention strategies, reduce churn, and support data-driven decision-making.

KPI SUMMARY

KPI	Value
Total Customers	64,374
Retained Customers	33,881
Churned Customers	30,493
Churn Rate	47.37%
Average Age	42 Years
Average Spend	₹541

Introduction :

Customer Churn Analysis is a business analytics project aimed at identifying customer retention patterns and understanding the factors that contribute to customer attrition. Customer churn occurs when customers discontinue their relationship with a company, resulting in reduced revenue and increased acquisition costs.

The objective of this project is to analyze customer demographics, subscription types, contract lengths, and spending behavior to identify churn trends and provide actionable recommendations for improving customer retention and business performance.

SQL Server was used for data analysis, and Power BI was utilized to develop an interactive dashboard that enables effective visualization and decision-making.

Dataset Overview :

The customer churn dataset consists of 64,374 customer records and contains demographic, behavioral, and subscription-related information. The dataset was used to analyze customer retention patterns and identify factors influencing customer attrition. The data includes customer demographics, subscription details, contract information, spending behavior, and churn status.

The major attributes included in the dataset are:

- CustomerID – Unique identifier for each customer.
- Age – Age of the customer.
- Gender – Customer gender information.
- Subscription Type – Type of subscription plan chosen by the customer.
- Contract Length – Duration of the customer contract.
- Total Spend – Total amount spent by the customer.
- Churn Status – Indicates whether the customer was retained or churned.

The dataset was imported into SQL Server for analysis and Power BI for visualization and dashboard development.

Customer Analysis :

Customer analysis provides insights into the overall customer base and demographic distribution. The dataset contains a total of 64,374 customers, representing a diverse customer population.

The average customer age was found to be approximately 42 years, indicating that the majority of customers belong to the middle-aged demographic segment. Gender analysis revealed that female customers slightly outnumber male customers.

Understanding customer demographics helps organizations identify target customer segments and develop customer-centric business strategies.

Churn Analysis

Customer churn analysis evaluates the percentage of customers who discontinue their relationship with the company. The analysis revealed a churn rate of 47.37%, indicating that nearly half of the customer base experienced attrition.

Out of 64,374 customers, approximately 30,493 customers churned, while 33,881 customers were successfully retained. This level of churn highlights the importance of customer retention programs and loyalty initiatives.

Identifying churn patterns enables businesses to take proactive measures to improve customer satisfaction and reduce customer loss.

Subscription Analysis

Subscription analysis was conducted to evaluate customer distribution across various subscription plans. The Standard subscription plan recorded the highest number of customers, followed closely by Basic and Premium plans.

The relatively balanced distribution of customers across subscription categories suggests that customers have diverse service preferences. However, churn patterns within these subscription types can provide valuable insights into customer satisfaction and service quality.

Analyzing subscription behavior allows businesses to optimize pricing strategies and improve subscription offerings.

Contract Analysis

Contract analysis helps evaluate customer preferences regarding contract duration. The analysis showed that Monthly contracts had the highest customer count, followed by Annual and Quarterly contracts.

Customers with shorter contract durations generally exhibit greater flexibility, while longer contracts may indicate stronger customer commitment and loyalty.

Understanding contract preferences helps organizations design retention strategies and develop suitable service plans that meet customer needs.

Customer Demographics Analysis

Customer demographic analysis provides insights into the composition of the customer base. Female customers accounted for approximately 34.35 thousand customers, while male customers represented approximately 30.02 thousand customers.

The relatively balanced gender distribution suggests that the company's products and services appeal to a broad customer audience.

Demographic analysis enables organizations to design targeted marketing campaigns and personalize customer experiences based on customer segments.

Business Impact

The findings from this customer churn analysis provide valuable insights that support strategic business decision-making. By understanding customer behavior, subscription preferences, and churn patterns, organizations can develop effective retention strategies and improve customer satisfaction.

The analysis enables businesses to identify high-risk customer groups, optimize subscription plans, improve service quality, and reduce customer attrition. These insights contribute to increased customer loyalty, improved profitability, and sustainable business growth.

Conclusion

The Customer Churn Analysis project successfully evaluated customer retention patterns and identified important factors influencing customer attrition. The analysis revealed a churn rate of 47.37%, with Standard subscriptions having the largest customer base and Monthly contracts being the most preferred contract type.

The results demonstrate the importance of customer analytics in understanding customer behavior and improving retention strategies. The interactive Power BI dashboard provides valuable insights into customer demographics, subscription preferences, and churn trends.

Overall, this project highlights the role of data analytics and business intelligence in supporting data-driven decision-making and enhancing customer relationship management.

Dashboard Visualization

Figure 1 illustrates the Customer Churn Analysis Dashboard developed using Power BI. The dashboard provides an interactive view of customer distribution, churn patterns, subscription analysis, contract preferences, demographic analysis, and key business insights.

The dashboard enables decision-makers to monitor customer behavior, identify churn trends, and make informed business decisions through data visualization and analytical reporting.

Figure 1: Customer Churn Analysis Dashboard

